DCS-Gait: A Class-Level Domain Adaptation Approach for Cross-Scene and Cross-State Gait Recognition Using Wi-Fi CSI

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***Abstract*— Wi-Fi CSI-based gait recognition is a non-intrusive passive biometric identification technology that has garnered significant attention in the fields of security and smart furniture due to its user-friendly nature. However, in practical application scenarios, gait recognition systems face the challenge of reliably identifying subjects across different scenes or states. To overcome this challenge, this paper proposes DCS-Gait, a domain adaptation solution for cross-scene and cross-state gait recognition based on Wi-Fi CSI. DCS-Gait leverages a novel data distribution measurement called Cross-Attention Metric to align the class-level data distribution differences, enabling the model to learn invariant features across scenes and states. To address the issue of data annotation, we employ a pre-training method to obtain pseudo labels for the dataset. Additionally, a combined matching filtering technique is utilized to generate high-quality pseudo labels for unrecognized data, which can be further employed for supervised model training. We evaluated the effectiveness of DCS-Gait on a large test set consisting of 34 subjects, 2 scenes, and 3 different states, and the results demonstrate significant improvements over the state-of-the-art baselines in both cross-scene and cross-state gait recognition tasks. DCS-Gait provides a promising and reliable solution for accurate cross-scene and cross-state gait recognition in real-world settings.**

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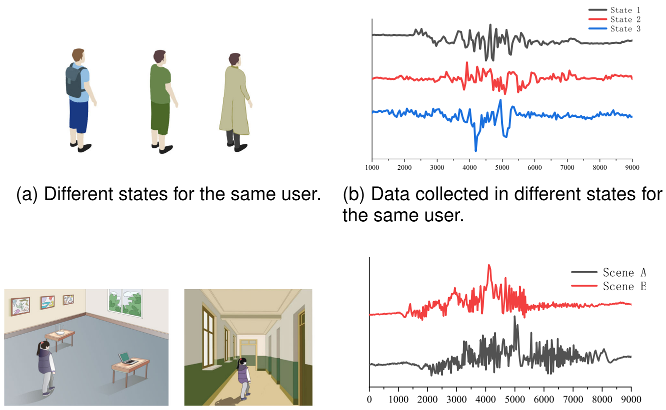
***Index Terms*— Wi-Fi CSI, gait recognition, cross-scene, cross-state, domain adaptation, cross-attention metric.**

I. Introduction

WITH the widespread use of pervasive computing, Unobtrusive User Identification (UUI) [1] has gained significant attention in research. Gait-based person identification effectively identifies individuals by leveraging the uniqueness of their gait [2]. Compared to biometrics based on acoustics [3], [4], face [5], [6], fingerprint [7], [8], and iris [9], [10], gait-based identification technology is well-suited for the task of Unobtrusive User Identification as it doesn't require active user participation. Currently, the most commonly used techniques for gait identification include image-based methods [11], [12], [13], wearable sensors [14], [15], [16], and RF signals [17], [18]. However, image-based methods are limited by environmental factors like lighting conditions and occlusions, which can also raise privacy concerns. Wearable sensors can be expensive and uncomfortable, making them unsuitable for noncooperative subjects. On the other hand, RF-based approaches are more versatile as they can penetrate obstacles and are not impacted by light or darkness. Additionally, the prevalence of wireless devices has made RF signals more accessible, reducing deployment costs. Previous studies have established a correlation between RF signals and walking patterns [19], [20], [21].

In recent years, the feasibility of gait recognition based on Wi-Fi signals has been explored. On the one hand, Wi-Fi frequency radio signals can penetrate clothing and bounce off the human body [22], providing stable human features. Wang et al. [23] developed the WiFiU system, which utilized a machine learning approach to achieve a recognition accuracy of up to 93% with a 50-person dataset. Akarsh Pokkunuru et al. [24] proposed the NeuralWave system, which utilized a deep learning approach and attained up to 89.9% recognition accuracy with a 24-person dataset.However, these methods require extensive training with previous examples of the same individuals walking in the same environment. On the other hand, Wi-Fi signals are heavily influenced by their surroundings, as they are susceptible to interference and environmental noise [25]. When the test and training environments differ, this can result in a model's failure to function. These challenges hinder the practical application of this technology, making it



(a) Different states for the same user. (b) Data collected in different states for the same user.

(c) Different scenes for the same user. (d) Data collected in different scenes for the same user.

Fig. 1. Illustration of the differences in gait data in different scenes or states. (a) and (b) demonstrate gait data with different attire, while (c) and (d) depict gait data in different scenes. The observed data disparities indicate significant changes in the data distribution when attire or environment varies. These variations may render previously trained models unable to adapt to domain shifts.

necessary to investigate cross-domain and robust recognition schemes.

However, the application of Wi-Fi signals to gait identification in realistic environments poses several challenges. Firstly, there is the challenge of accurately identifying cross-state or cross-scene gait. The signal propagation can be significantly perturbed by the user's state or scene, such as wearing different clothing or in different scenes, which leads to a shift in data distribution, as illustrated in Fig.1. Failure to account for the difference between the target domain (state or scene) and the source domain (original state) can severely impact the model's performance. Secondly, there are limitations associated with annotated datasets. Most Wi-Fi CSI-based gait identification methods rely on supervised approaches that require extensive and time-consuming dataset annotation. To address this issue, Fan et al. [26] proposed combining video-based approaches to reduce the annotation requirements. However, this solution increases deployment costs and raises privacy concerns, limiting its practical applicability. Furthermore, capturing scenes poses its own limitations. Collecting large-scale datasets from various environments is both time-consuming and challenging due to the high correlation between RF signals and the environment. This, in turn, can make it difficult to identify a suspect who attempts to disguise themselves across different scenes. For example, if a suspect appears in Scene A and later disguises themselves to appear in Scene B, determining whether they are the same person becomes a challenge.

In order to overcome the aforementioned limitations of gait identification based on Wi-Fi signals, this paper presents a novel approach for cross-state and cross-scene gait identification via unsupervised domain adaptive using Wi-Fi Channel State Information (CSI), as shown in Fig.2. Our method called DCS-Gait, implements a new data distribution metric, cross-attention distance, to achieve class-aware condition alignment at the class level. This design effectively learns the invariant features from both the original and current states simultaneously. To tackle the difficulty of annotating new state datasets, we use pre-training to obtain pseudo-labels for new gait data and matching filtering to generate high-quality labeled training pairs, thus achieving unsupervised training of gait data. In summary, these techniques allow the model to efficiently learn and classify artifactual gait data without the need for extensive manual annotation.

To assess the effectiveness of our approach, we constructed a comprehensive test set comprising 34 subjects, 2 scenes, and 6 different costumes, allowing for an extensive evaluation of system performance. The proposed method holds potential for diverse applications, including security surveillance (e.g., precise identification of suspects attempting disguise in non-crime scenes) and smart homes (e.g., resident identification based on different areas and customized services derived from it).

Our contributions can be summarized as follows:

1) To the best of our knowledge, this is the first exploration of utilizing Wi-Fi CSI for gait recognition across different domains. We propose a novel model based on class-level unsupervised domain adaptation. The model employs a cross-attention metric to calculate the differences in class-level data distributions. This design explicitly establishes domain differences at the class level, achieving alignment based on class-level perception conditions and addressing the challenges of cross-scene and cross-state gait recognition.
2) Our approach tackles the challenge of data annotation in unsupervised learning by employing a pre-training method. Through this method, we generate pseudo-labels for the dataset, effectively addressing the scarcity of labeled data. Furthermore, we enhance the quality of these pseudo-labels by utilizing a combined matching filtering technique, particularly for unrecognized data. As a result, we ensure the production of high-quality pseudo-labels, providing reliable labeled data that can be used for subsequent supervised model training.
3) To evaluate the performance of CSGait, we conducted extensive experiments using a comprehensive test set consisting of 34 subjects, 2 scenarios, and 3 different dressing codes. The experimental results demonstrate significant achievements, achieving a recognition accuracy of 94.61% in the cross-state gait recognition task. This represents a substantial improvement of approximately 20% compared to the baseline method that lacks data distribution optimization. Moreover, in the cross-scene gait recognition task, our method outperforms baseline approaches by approximately 50%.

The rest of the paper is organized as follows. Section II provides a brief review of related work. Section III details the proposed method. Section IV presents the experiments and discussions. Finally, we conclude the paper in Section V.

II. Related Works

A. Gait Identification

In recent years, several studies have investigated the use of Wi-Fi CSI-based methods for person identification. Existing gait recognition techniques based on Wi-Fi CSI can be

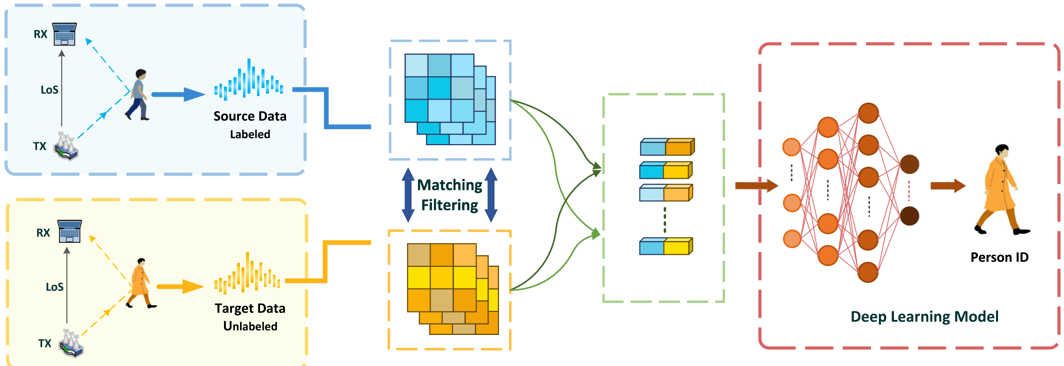


Fig. 2. Illustration of our proposed approach. As the same user passes through the Wi-Fi sensing area wearing different clothes (state) in different scenes, we utilize an attention mechanism and class-aware alignment matching algorithm to extract stable identifiable features from the obtained data, thereby achieving accurate recognition of gait.

categorized into two groups: manual feature extraction-based methods and deep learning-based methods, depending on the approach used for feature extraction.

The manual feature extraction-based approach selects statistical features such as signal waveforms from the received CSI data based on a priori knowledge and combines them together as a feature matrix for further system training and recognition. Zhang et al. [27] proposed Gate-ID by selecting mean, maximum, minimum, skewness, kurtosis, variance, and mean cross-rate features of CSI waveforms and using frequency domain features such as normalized entropy, normalized energy, and FFT peaks to finally achieve gait recognition. Hong et al. [28] proposed WFID by analyzing the CSI measured subcarrier amplitude frequency (SAF) to finally achieve user identification. Xin et al. [29] proposed FreeSense uses principal component analysis to extract the gait features of different users for user identification purposes.

Deep learning-based approaches use deep learning networks to learn features directly from the received CSI samples to construct feature matrices without prior knowledge. Lin et al. [30] combined CNN and ResNet to build their system model, where CNN downsamples the original data, and then ResNet extracts high-level features from the CSI samples for final gait recognition. Wang et al. [31] proposed CSIID using a combination of CNN and long short-term memory (LSTM) to extract gait features for gait recognition. In HumanFi proposed by Ming et al. [32], a recurrent neural network LSTM is used to learn human gait features to identify different users. Wihi proposed by Ding et al. [33] uses a recurrent neural network model to study the power and energy distribution of CSI data to finally achieve gait recognition

However, these existing methods have limitations as they rely on previous instances of the same person walking in the same state and scene, and they require a large number of labeled samples. These methods do not effectively solve the challenge of cross-state and cross-scene gait recognition.

B. Unsupervised Domain Adaptation

Depending on whether the target domain is labeled in training, domain adaptation can be classified as supervised domain adaptation, semi-supervised domain adaptation, and unsupervised domain adaptation. Among them, unsupervised domain adaptation is the most challenging task because of its inability to gain the target data labels. The unsupervised domain adaptation mainly solves the new target domain data distribution offset problem by domain-level and category-level.

1) Domain-Level: The goal of unsupervised domain adaptation is to optimize all data in the source and target domains to the same distribution, thus achieving domain adaptation. For example, Long et al. [34] use MMD as an explicit measure of the difference in source-target distribution in migration learning, and reduce the difference in distribution between source and target data by decreasing the MMD value. Sun and Saenko [35] align the second-order statistics of source and target domains by a linear transformation to reduce the difference in distribution between source and target data, and achieve domain-level unsupervised domain adaptation.

2) Category-Level: The unsupervised domain adaptation of category-level is optimized by fine-graining the distribution of the source and target domains corresponding to the category data. Obviously, fine-grained alignment enables more accurate distribution alignment of the same category data. For example, Yang et al. [17] propose a G-SAC method that overcomes the class-level data distribution bias and finally accomplishes the unsupervised domain adaption task requirements effectively. Kang et al. [36] propose a Contrastive Adaptation Network (CAN), which explicitly models the intra-class domain discrepancy and the inter-class domain discrepancy, through an alternating update strategy for training CAN and implemented Unsupervised Domain Adaptation.

Most of these studies have made significant progress in unsupervised and adaptive image-based approaches, but they are not suitable for our Wi-Fi CSI-based gait data. This is due to the fact that wireless data contains less information compared to image data and is more sensitive to environmental factors, resulting in noticeable differences in data distribution. In this paper, we propose a class-level data distribution discrepancy metric to achieve class-aware condition alignment. This novel approach enables the model to simultaneously learn invariant features from different domains, effectively addressing the challenge of cross-scene and cross-state identification.

III. Methods

The proposed DCS-Gait framework consists of feature extractor, classifier and feature alignment modules.

The framework takes as input the training data pairs obtained through the pseudo label matching filtering mechanism. The source domain data and target domain data in the training data pairs are subjected to feature extraction separately. The classification is trained using softmax cross-entropy loss. There is a feature alignment module between the source domain branch and the target domain branch. This module receives inputs from both branches and enforces the extraction of domain-shared identity features by limiting the distributional differences between the branches. This ultimately enables gait recognition in different scenarios and states.

This section presents the DCS-Gait framework. Firstly, we provide a concise explanation of the domain concept in the context of cross-sense and cross-state gait person identification. Subsequently, in the following subsections, we elaborate on the pseudo labeling and cross-attention metric in detail. The overall architecture of the DCS-Gait is shown in Fig.3.

A. Preliminary

In this paper, the data contains two parts: the source domain and the target domain. The gait data with category labels constitute the source domain data, and new state or scene gait data(no labels) constitute the target domain data. Both source and target domains have the same feature space and label space, while the data probability distributions are different. A sample in the source domain data contains input data x and output labels y, and its data probability distribution is P_s, the source domain data can be represented as $\mathcal{D}_s = \{\mathcal{X}_s, \mathcal{Y}_s, P_s\}$, where $\mathcal{X}_s$, $\mathcal{Y}_s$ are the feature space of the input data and the label space of the output labels, respectively. A sample in the target domain data contains only input data, and its data probability distribution is P_t, the target domain data can be represented as $\mathcal{D}_t = \{\mathcal{X}_t, P_t\}$. The goal of unsupervised domain adaptation is to use the source domain data to learn a prediction function on the target domain $f : x_t \mapsto y_t$ such that f has the minimum prediction error on the target domain when the feature space and label space are the same i.e. $\mathcal{X}_s = \mathcal{X}_t$, $\mathcal{Y}_s = \mathcal{Y}_t$, but the data probability distribution is different.

$$f^* = \arg\min_f \mathbb{E}_{(x,y)\in\mathcal{D}_t} \Gamma(f(x), y). \tag{1}$$

Our approach involves employing neural networks to bridge the gap between the distributions of the source and target domains. This allows us to address the challenge of lacking classification models trained on the original data when new state or scene gait data is introduced. By leveraging neural networks, we aim to facilitate the adaptation of the model to the new data and enable accurate classification in the target domain.

B. Target Domain Pseudo-Labeling and Matching Filtering

Considering class-level information is crucial in domain adaptation to prevent the accumulation of generalization errors from the source domain [37]. In this study, we address this issue by adopting a comprehensive approach that operates at both the domain level and the class level. Our approach involves three key steps: Firstly, we employ pre-training to obtain pseudo-labels for the target domain data, facilitating subsequent adaptation. Secondly, through matching filtering, we construct pairs of samples from the same class in the source domain, creating high-quality labeled training data. Finally, leveraging class-level unsupervised domain adaptation, we bridge the gap between domains, ensuring effective adaptation and improving overall performance. This holistic approach enables us to overcome the challenge of incorporating new state or scene gait data when classification models trained on the original data are not available.

Pseudo-labeling Drawing inspiration from [38], we first transform the target domain data using a feature extractor (which has been pre-trained using source domain data) to obtain its probability distributions δ. These distributions can be utilized to calculate the initial center of mass for each category in the target domain by using weighted k-means clustering:

$$c_k^{(0)} = \frac{\sum_{x_t\in\mathcal{X}_t} \delta_k(F_{x_t}) F_{x_t}}{\sum_{x_t\in\mathcal{X}_t} \delta_k(F_{x_t})} \tag{2}$$

where $\delta_k(F_{x_t}) = \frac{\exp(F_{x_{t\_k}})}{\sum \exp(F_{x_t})}$ denotes the probability distribution of the target domain data x_t in class k. F_{x_t} means the features (output of the Feature Extractor) of the target domain data x_t. $F_{x_{s\_k}}$ denotes the features of the samples in the target domain data x_t which are predicted as class k by the pre-training model. The pseudo-label of the target data can be generated by the nearest neighbor classifier:

$$\hat{y}_t = \arg\min_k d(c_k^{(0)}, F_{x_t}) \tag{3}$$

where $d(c_k^{(0)}, F_{x_t})$ is the distance between $c_k^{(0)}$ and F_{x_t}. Based on the pseudo label, we can calculate the new center and the pseudo label:

$$c_k^{(1)} = \frac{\sum_{x_t\in\mathcal{X}_t} \mathbb{I}(\hat{y}_t = k) F_{x_t}}{\sum_{x_t\in\mathcal{X}_t} \mathbb{I}(\hat{y}_t = k)} \tag{4}$$

$$\hat{y}_t = \arg\min_k d(c_k^{(1)}, F_{x_t}) \tag{5}$$

Here, $\mathbb{I}(\hat{y}_t = k)$ denotes the probability distribution of the $\hat{y}_t$ in class k. It is worth noting that Equations (4) and (5) is a new iteration of Equations (2) and (3) which are calculated in the same way.

The pseudo-labeling of the target domain can be done by repeating the above equation.

1) Matching Filtering: The noise in the acquired pseudo-labels may cause a significant skew in the data distribution, and the unsupervised generation of pseudo-labels for the target domain data may result in an imbalance in the sample count between the source and target domains. To address these challenges, we propose a feature matching approach for pseudo-label filtering. Specifically, we match data of the same class from both the source and target domains, as represented by:

$$\mathbb{D}_S = \{(x_{s\_k}, x_{t\_k}) | x_{t\_k} \in \mathcal{X}_t(\min d(F_{x_{s\_k}}, F_{x_{t\_k}}))\} \tag{6}$$

In the matching filtering process, x_{s_k} and x_{t_k} denote the samples of class k in the source and target domains, respectively. The $d(F_{x_{s\_k}}, F_{x_{t\_k}})$ denote the cosine distance between the features $F_{x_{s\_k}}$ and $F_{x_{t\_k}}$. $\mathcal{X}_s(\min d(F_{x_{t\_k}}, F_{x_{s\_k}}))$

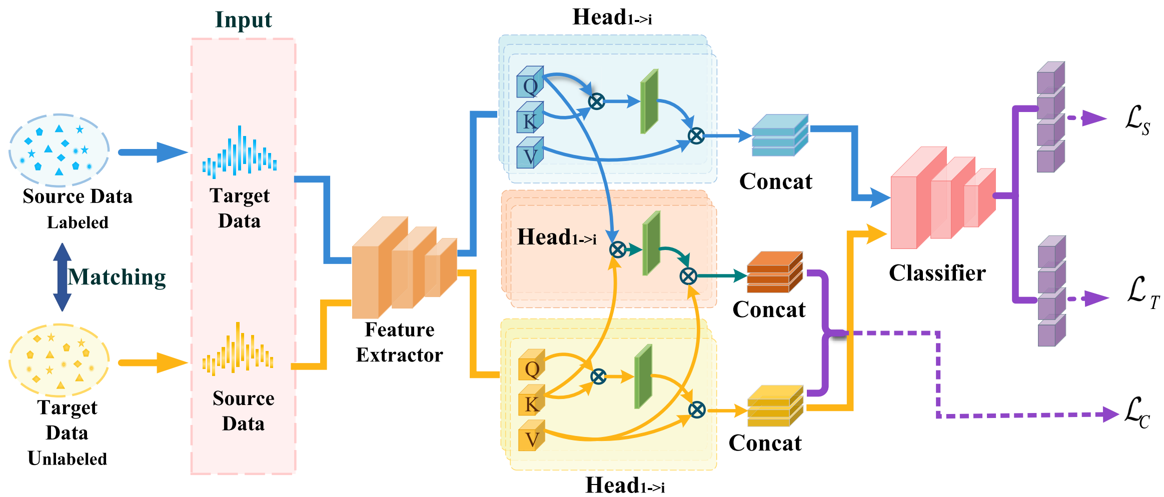


Fig. 3. The proposed DCS-Gait framework. DCS-Gait consists of three modules: a Feature Extractor, a Classifier and a Feature Alignment Module. The framework takes training data pairs, obtained through a pseudo label matching filtering mechanism, as input. In the training data pair, source and target domain data undergo feature extraction through the blue and orange branches, respectively. Classification is performed using softmax cross-entropy loss. The Feature Alignment Module, positioned between the source and target domain branches, utilizes inputs from both branches. It compels the feature extractor to extract domain-shared identity features by constraining the distributional difference between the Feature Alignment Module branches and the target domain branches. This mechanism facilitates cross-scene and cross-state gait recognition.

denote the data that has the smallest cosine distance from the source domain data x_{s_k} in the target domain data $\mathcal{X}_t$. It's worth mentioning that the number of matches between the source and target domain data is not limited, potentially resulting in many-to-one relationships, thus allowing for the utilization of high-confidence pseudo-labels. To further expand the training data pairs, the same process is applied to the target domain data, where the source domain data is used as the category label of the training data pair, considering the real label of the source domain data to be more reliable than the pseudo-labels in the target domain. This can be expressed as follows:

$$\mathbb{D}_T = \{(x_{t\_k}, x_{s\_k}) | x_{s\_k} \in \mathcal{X}_s(\min d(F_{x_{t\_k}}, F_{x_{s\_k}})\} \tag{7}$$

The final training data pair is $\mathbb{D} = \{\mathbb{D}_S \cup \mathbb{D}_T\}$.

C. Cross-Attention Metric

The utilization of attention mechanisms is widely recognized for their ability to efficiently integrate temporal information in sequences [24]. This is particularly crucial for the extraction of gait information from CSI sequences, and it can be described as follows:

$$\mathcal{F}(Q, K, V) = soft\max(\frac{QK^T}{\sqrt{d_k}})V \tag{8}$$

The parameters Q, K, and V can be obtained by mapping the input F. The dimension of the Q and K matrices is denoted as d_k, and $\sqrt{d_k}$ is a scaling factor that facilitates smoother gradients. The output of the self-attention is calculated as a weighted sum. In order to measure the distribution difference between the source and target domains, we use the cross-attention metric, as inspired by [39]. Specifically, we calculate a weight parameter for sample similarity by matching the source domain Q_S with the target domain K_T. This weight parameter is used to selectively extract the information V_T of the target domain, resulting in $\mathcal{F}_C$. The distribution difference is transformed from the source domain to the difference between $\mathcal{F}_C$ and $\mathcal{F}_T$, and the alignment of the source and target domains is achieved by minimizing the distance between $\mathcal{F}_C$ and $\mathcal{F}_T$.

$$\mathcal{F}_C(Q_S, K_T, V_T) = soft\max(\frac{Q_S {K_T}^T}{\sqrt{d_k}})V_T \tag{9}$$

where Q_S is the query from the source domain data and K_T, V_T is the key and value from the target domain. The difference in distribution between $\mathcal{F}_C$ and $\mathcal{F}_T$ can be quantified as:

$$\mathcal{L}_c = \sum \mathcal{F}_C \log \mathcal{F}_T \tag{10}$$

D. Objective and Training

The final optimized loss function is obtained in the above way as follows:

$$\mathcal{L} = \mathcal{L}_S + \mathcal{L}_T + \mathcal{L}_C \tag{11}$$

where $\mathcal{L}_C$ is the loss of cross-attention metric, $\mathcal{L}_S$, $\mathcal{L}_T$ is the classification loss of source and target domains, the loss is counted using the cross-entropy function:

$$\mathcal{L}_s = -\frac{1}{N_s} \sum_{i=1}^{N_s} \sum_{c=1}^{k} y_s \log(\hat{y}_s) \tag{12}$$

$$\mathcal{L}_t = -\frac{\alpha}{N_s} \sum_{i=1}^{N_s} \sum_{c=1}^{k} y_s \log(\hat{y}_t) \tag{13}$$

where α is the weight parameter.

It is worth noting that the data in both the source and target domains are supervised using the labels of the training data $\mathbb{D}$. The introduction of $\mathcal{L}_t$ can further generalize the model and improve the recognition accuracy.

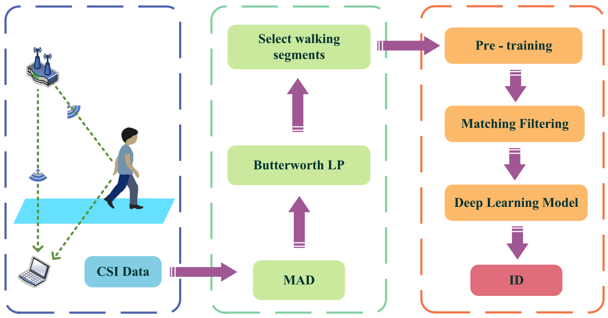


Fig. 4. System Workflow. The system uses a pair of commercial WiFi devices for data collection. Subsequently, the absolute median difference (MAD) method is used to filter out additive and hardware noise in the environment; a Butterworth low-pass filter is used to filter out high-frequency noise; and a non-overlapping sliding window method is used to select out walking segments. Then pre-training is used to obtain pseudo-labels, matching filtering is used to obtain high-quality training data pairs with labels, and finally, a deep learning model (DCS-Gait) is used to achieve unsupervised gait identification.

IV. Experiments

To evaluate the effectiveness of our approach, we conducted experimental validation in real-world scenarios. The system workflow, as depicted in Fig.4, consists of three main components: data acquisition, data pre-processing, and a deep learning module. The system takes Wi-Fi signals as input and extracts body state features that are independent of attire and scene for identification purposes. By incorporating an attention mechanism and a class-aware alignment algorithm, the deep learning module achieves accurate recognition of gait patterns across different scenes and states.

A. Implementation

In our experiments, the horizontal distance between the sender and receiver was 2.5 meters, and the height of the devices from the ground was 0.8 meters (this height allowed movements with significant identity features, such as arm and leg swings, to be captured effectively). We considered three states in two scenes in the measurements: normal dress state in the hall (H_n), backpack-loaded state in the hall (H_p), long trench coat covering the lower legs in the hall (H_c), normal dress state in the corridor (C_n), backpack-loaded state in the corridor (C_p), and long trench coat covering the lower legs in the corridor (C_c). Wi-Fi gait data was collected from a total of 34 subjects, including 26 males and 8 females. The subjects walked a distance of 7 meters, and the signal acquisition duration was approximately 7-9 seconds. Each participant performed a single gait pattern 30 times, resulting in a total of 6120 data points collected for (34 subjects) $\times$ (2 scenes) $\times$ (3 patterns) $\times$ (30 repetitions). In each task, the training dataset comprises all source domain data (34 subjects $\times$ 1 patterns $\times$ 30 repetitions) and 80% of the target domain data (34 subjects $\times$ 1 patterns $\times$ 24 repetitions). The test dataset consists of the remaining 20% of the target domain data (34 subjects $\times$ 1 patterns $\times$ 6 repetitions), which was not used in the training process. The deep learning model was implemented on the PyTorch framework.

B. Data Pre-Processing

As the collected CSI data contains a large amount of noise and abnormal values, it will affect the final gait recognition results. Therefore pre-processing (noise reduction and filtering, etc.) of the signal is necessary.

The noise present in CSI data can be categorized into two types: internal noise and external noise, based on their sources. Internal noise originates from hardware components in the transceiver, such as Cyclic Shift Diversity (CSD), Sampling Time Offset (STO), and Sampling Frequency Offset (SFO) [40]. This type of noise typically manifests as phase shifts in the data. Eliminating internal noise requires precise time synchronization at the transceiver to achieve phase alignment, which often necessitates modifications to common commercial Wi-Fi equipment. To mitigate the impact of internal noise, this paper focuses on using the amplitude of the CSI signal for gait identification. On the other hand, external noise primarily arises during signal propagation and is mainly in the form of additive noise. This type of noise is inevitable in signal acquisition.

Therefore, we employed the sliding window Median Absolute Deviation (MAD) method to mitigate additive noise. MAD serves as an anomaly detection technique by computing the sum of distances between all data points and the median value. It is calculated using Equations (14) and (15), as shown at the bottom of the page, where $X(t,n)$ represents the signal amplitude of the CSI in the n-th window at time t, $X'(t,n)$ is the processed CSI signal amplitude, $X_{median}(n)$ is the median value of the n-th window, and N is the threshold. This method effectively eliminates additive noise.

Furthermore, considering that the human gait information is primarily mapped within the frequency range of 0-32Hz [23], a Butterworth low-pass filter with a passband frequency of 32Hz is applied to eliminate high-frequency noise from the CSI data.

After the aforementioned preprocessing, the noise-reduced CSI signal is obtained. Nevertheless, the preprocessed CSI data cannot be directly utilized for identification. This is because the collected CSI fragments encompass the entire user walking process, necessitating distinct methods for efficient feature extraction to identify different states. Consequently, fragment detection assumes a crucial role.

$$MAD(n) = median(|X(t,n) - median(X(t,n))|) \tag{14}$$

$$X'(t,n) = \begin{cases} X_{median}(n) + N \times MAD(n), & X(t,n) > X_{median}(n) + N \times MAD(n) \\ X_{median}(n) - N \times MAD(n), & X(t,n) < X_{median}(n) - N \times MAD(n) \\ X(t,n), & X_{median} - N \times MAD(n) < X(t,n) < X_{median}(n) + N \times MAD(n) \end{cases} \tag{15}$$

TABLE I
CROSS-STATE IDENTIFICATION ACCURACY (%) FOR DIFFERENT ADAPTATION TASKS

	H_n→H_c	H_n→H_p	H_c→H_n	H_c→H_p	H_p→H_n	H_p→H_c
Source-only	75.98	73.53	70.59	78.43	71.63	79.11
BNM [34]	78.82	77.21	76.32	80.26	73.89	79.84
DAAN [41]	86.7	86.02	79.07	86.75	82.17	81.88
DAN [42]	82.27	79.96	81.73	84.48	84.95	81.68
DANN [43]	87.71	81.66	85.02	86.96	80.12	86.38
DeepCoral [35]	82.06	82.31	83.79	85.38	82.87	79.45
DSAN [44]	87.68	81.73	83.27	89.54	79.75	81.03
DTA [45]	76.66	74.28	75.96	79.31	71.02	78.12
Ours	**90.68**	**90.69**	**94.61**	**91.66**	**92.64**	**92.16**

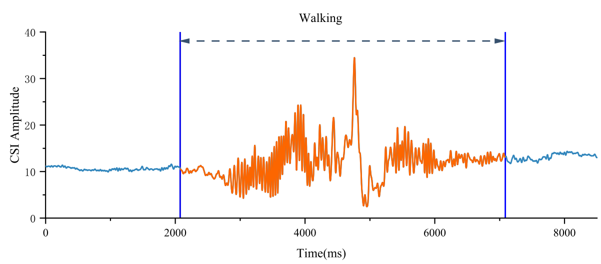


Fig. 5. The pre-processed signal consists of a walk fragment, depicted in red, obtained through interception, and a redundant fragment, represented in blue.

From the CSI signal, it is discernible that when no user is walking, the CSI signal remains in a stable state. In the presence of users, the CSI exhibits significant fluctuations. To discern these variations, the system employs the non-overlapping sliding window variance method to detect segment volatility based on the conspicuous difference between dynamic and static CSI signals. The starting and ending points of the movement phase are determined by calculating the variance of the window and setting the corresponding threshold. This process is expressed by Equations (16) and (17), as shown at the bottom of the page. Where $G_s(t_i, n)$ is the i^{th} value of the t^{th} windows of the n^{th} subcarriers, T_w is the total number of packets in each sliding windows, $\overline{G_s(t, n)}$ is mean of the t^{th} windows of the n^{th} subcarriers of $G_s(t_i, n)$. ϕ_1 is the threshold of candidate points, preliminarily determining the start and end of the candidate movement phase point. The threshold ϕ_2 added to the buffer mechanism is used to conclusively determine the start and end points of the final movement phase.

The processed CSI signal is depicted in Fig.5 as an illustration of the data after undergoing the aforementioned processing steps.

C. Performance Evaluation

We evaluated the DCS-Gait method on the tasks of cross-state gait recognition, cross-scene gait recognition, and cross-state and cross-scene gait recognition. To ensure an objective assessment of the effectiveness of our proposed method, we compared it with several state-of-the-art unsupervised domain adaptation algorithms, including BNM [34], DAAN [41], DAN [42], DANN [43], DeepCoral [35], DSAN [44], and DTA [45].

BNM [34] utilizes batch normalization to improve the recognition capability of target domain data. DAAN [41] addresses domain mismatch and enhances transfer learning accuracy by using dynamic distribution. DAN [42] measures the differences between source and target domain distributions and minimizes them to reduce disparities. DANN [43] employs Generative Adversarial Networks to learn domain-invariant features. DeepCoral [35] aligns the second-order statistical information of source and target domains to reduce data distribution differences. DSAN [44] proposes a flexible and effective soft-labeling mechanism for category matching. DTA [45]uses adversarial dropout to learn discriminative features and adapt the feature distribution to target domain data.

To maintain fairness and better align with our collected gait data, all of these domain adaptation algorithms utilize the same feature extractor, a 14-layer convolutional neural network, as well as the same classifier, a 3-layer perceptron. This standardized setup enables a reliable and meaningful comparison of our method with the aforementioned approaches.

1) Cross-State Gait Identification: In the experiment, we evaluated the performance of our proposed method in recognizing gait across different states. We selected three states (H_n, H_c, and H_p) from the hall and randomly assigned two of them as the source and target domains. The results are shown in Table I. We compared DCS-Gait with the Source-only method, which lacks domain adaptation, and seven other state-of-the-art domain adaptation methods. The results demonstrate that our method consistently outperformed the Source-only method (by a minimum of 13.05% and up to 24.02%) and other domain adaptation methods in terms of accuracy. Importantly, our method achieved an impressive accuracy of up to 94.61% in cross-state gait identification, highlighting its robustness and competitiveness in this task.

$$Var(t) = \frac{1}{NT_W}\sum_{n=1}^{N}\sum_{i=1}^{T_w}\left(G_s(t_i, n) - \overline{G_s(t, n)}\right)^2 \tag{16}$$

$$\begin{cases} X = begin, Var(t:t+2) > \phi_1 \& Var(t:t+4) > \phi_2 \& Var(t-4:t) < \phi_2 \\ X = end, Var(t:t+2) < \phi_1 \& Var(t:t+4) < \phi_2 \& Var(t-4:t) > \phi_2 \end{cases} \tag{17}$$

TABLE II
CROSS-SCENE IDENTIFICATION ACCURACY (%) FOR DIFFERENT ADAPTATION TASKS

	H_n→C_n	C_n→H_n	H_c→C_c	C_c→H_c	H_p→C_p	C_p→H_p
Source-only	14.41	15.88	14.66	15.73	13.68	14.21
BNM [34]	26.81	25.87	21.72	20.67	20.91	28.78
DAAN [41]	26.93	26.71	24.89	28.05	26.69	25.84
DAN [42]	29.61	26.86	29.02	23.95	29.72	27.07
DANN [43]	28.04	22.53	28.46	23.07	24.59	28.27
DeepCoral [35]	23.47	28.50	24.48	22.40	27.62	22.29
DSAN [44]	24.71	21.94	21.60	27.92	23.94	25.13
DTA [45]	20.44	27.15	25.53	19.66	26.97	22.35
Ours	**59.31**	**67.65**	**61.07**	**69.92**	**62.41**	**66.37**

TABLE III
CROSS-STATE AND CROSS-SCENE IDENTIFICATION ACCURACY (%) FOR DIFFERENT ADAPTATION TASKS

	H_n→C_c	H_n→C_p	H_c→C_n	H_c→C_p	H_p→C_n	H_p→C_c
Source-only	12.94	13.43	12.90	12.53	13.48	11.54
BNM [34]	28.49	22.19	28.43	24.72	21.41	24.95
DAAN [41]	22.53	26.81	23.88	24.42	29.79	25.84
DAN [42]	21.86	26.47	21.15	23.94	26.34	24.45
DANN [43]	30.94	28.50	30.04	24.25	26.19	24.15
DeepCoral [35]	27.09	29.44	28.09	26.34	29.18	26.59
DSAN [44]	21.43	30.51	20.86	25.27	25.19	22.42
DTA [45]	26.30	26.28	23.22	26.64	24.24	25.48
Ours	**54.78**	**57.35**	**59.07**	**60.87**	**58.33**	**57.56**

2) Cross-Scene Gait Identification: To scientifically evaluate the performance of our proposed method in cross-scene gait identification, we selected gait data from one scene as the source domain and gait data from another scene as the target domain. The experimental results are shown in Table II. Similar to the previous cross-state experiments, we compared our method with the source-only method and seven other state-of-the-art domain adaptation methods. The results demonstrate a significant advantage in accuracy for our method, with an improvement of at least 40% compared to the source-only method. Additionally, it achieves a twofold increase in recognition rate compared to other domain adaptation methods. It is worth noting that there is a significant distribution shift between the source and target domains due to scene changes. Traditional domain adaptation methods mainly focus on pixel-level image domain adaptation, which may not be suitable for Wi-Fi signals that are heavily influenced by environmental factors. In contrast, our method is based on the Transformer architecture [46], widely used in natural language processing tasks, making it more suitable for handling gait data. The experimental results further validate the effectiveness of our approach.

3) Cross-State and Cross-Scene Gait Identification: In addition, we also address the more complex task of recognizing gait across different scenes and states, where both the scene and state change simultaneously. This scenario presents a meaningful and challenging situation for gait recognition. The experimental results are summarized in Table III. It is observed that our method achieves a significant improvement in recognition accuracy, with an increase of approximately 50% compared to the source-only method and nearly 30% compared to other domain adaptation methods. This highlights the effectiveness of our approach in mitigating the impact of scene and state changes on gait recognition.

By comparing the three different cross-domain scenarios, it can be observed that our method has significant advantages.

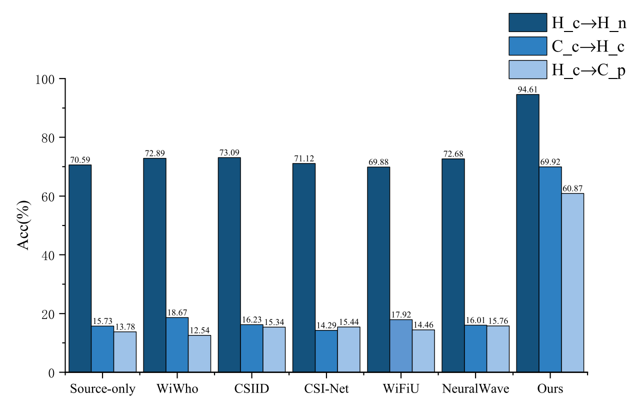


Fig. 6. Comparisons results with existing methods.

This is because our proposed novel data distribution metric effectively aligns the class-level data distribution differences, enabling the model to learn invariant features across scenes and states. As a result, DCS-Gait performs exceptionally well in all the aforementioned recognition tasks.

D. Comparisons With Existing Gait Recognition Methods

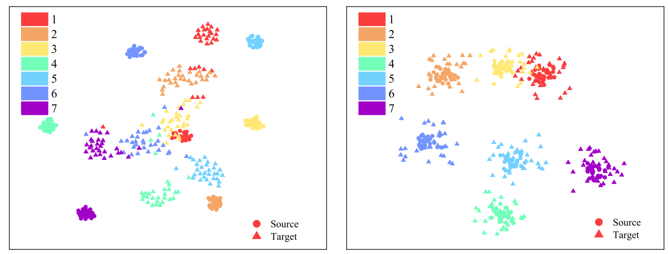
To demonstrate the advantages of DCS-Gait, we compared it with five state-of-the-art gait recognition methods: WiWho [47], CSIID [31], CSI-Net [18], WiFiU [23], and NeuralWave [24].

The experimental results are described in Fig. 6. Specifically, for the cross-state gait recognition task, we designate H_c as the source domain and H_n as the target domain. Similarly, for the cross-scene gait recognition task, we utilize C_c as the source domain and H_c as the target domain. In the cross-scene and cross-state gait recognition task, we use H_c as the source domain and C_p as the target domain. The experimental results demonstrate that existing gait recognition

TABLE IV
CLASSIFICATION ACCURACY OF ABLATION EXPERIMENTS (%)

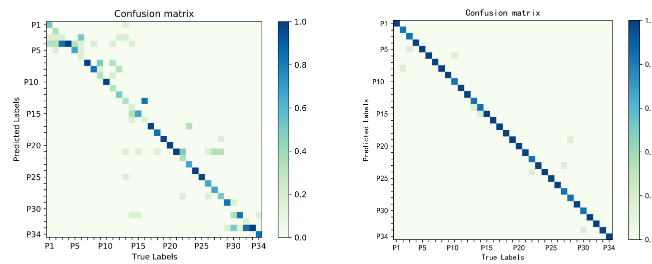
	H_n→H_c	H_n→H_p	H_c→H_n	H_c→H_p	H_p→H_n	H_p→H_c
Source-only	75.98	73.53	70.59	78.43	71.63	79.11
Ours(w/o PL&MF)	82.87	79.27	81.98	83.24	76.08	81.09
Ours(w/o C-A metric)	85.67	86.09	79.66	81.08	84.38	86.29
Ours	**90.68**	**90.69**	**94.61**	**91.66**	**92.64**	**92.16**

approaches fail to address the issue of significant recognition rate degradation caused by changes in state or scene. This is because when the domain changes, the data distribution undergoes noticeable variations, rendering the previously trained models inadequate for the new data distribution, resulting in decreased recognition accuracy. In contrast, DCS-Gait leverages pseudo-label matching filtering to generate highly confident data pairs and employs cross-attention metrics to effectively adapt to changes in data distribution, enabling the model to capture invariant features across domain transitions. DCS-Gait outperforms other methods in cross-domain, cross-state, cross-scene, and cross-state-and-scene gait recognition tasks.



(a) Source-only (b) Ours

Fig. 7. The t-SNE visualization of embedded features.



(a) Source-only (b) Ours

Fig. 8. The target domain confusion matrices.

E. Ablation Studies

1) Effectiveness of Pseudo-Labeling and Matching Filtering: In order to evaluate the influence of pseudo-labeling and matching filtering on the target domain, we conducted experiments where we randomly combined the source and target domains to create training data pairs. The results, presented in Table IV, indicate that the model's ability to adapt to the target domain at the class level was significantly compromised when pseudo-labeling and matching filtering were not utilized, and when the $\mathcal{L}_T$ component was removed. This degradation in performance underscores the effectiveness of these operations in improving domain adaptation.

2) Effectiveness of Cross-Attention Metric: To further validate the impact of the cross-attention metric in the class-aware setting, we excluded the $\mathcal{L}_C$ component from the model. The results, shown in Table IV, demonstrate that the cross-attention metric plays a crucial role in addressing the shift in data distribution caused by changes in state or scene. It effectively overcomes the limitations of the model in such scenarios and helps maintain high performance.

F. Experimental Results Visualization

To further demonstrate the effectiveness of our proposed method, we used t-SNE [48] to visualize the extracted features. As shown in Fig. 7, we selected 7 users, using H_c as the source domain and H_n as the target domain. Fig. 7a depicts the target domain data distribution as obviously shifted and with no clear class boundaries, leading to unsatisfactory performance in gait identification. Fig. 7b shows the improved data distribution after optimization, where the target domain data distribution has less shift and clear class boundaries, resulting in significantly improved identification accuracy.

In addition, we also present the results in the form of a confusion matrix. Fig. 8a shows the Source-only results, where there is clear confusion between all targets. Fig. 8b shows the results using our proposed method, with significantly improved identification accuracy. By comparing the two figures, it is clear that our proposed method is capable of addressing the task of cross-state and cross-scene gait identification.

Through visualizing the experimental results, DCS-Gait showcases class-aware conditional alignment at the class level, leading to a significant increase in inter-class distance and effectively addressing data distribution discrepancies. This design allows the model to learn invariant features during domain transitions, resulting in an outstanding performance in cross-scene and cross-state recognition tasks.

V. CONCLUSION

This paper introduces DCS-Gait, a novel approach for unsupervised gait identification across different states and scenes using Wi-Fi CSI signals. It effectively addresses the challenges faced by traditional models in cross-scene and cross-state gait recognition tasks. DCS-Gait offers the advantage of reducing deployment costs and enabling unsupervised gait recognition without extensive manual labeling. The approach involves projecting normal gait data from the source domain and gait data from the target domain (cross-state or cross-scene) into a feature space and aligning their class-level data distributions using a novel cross-attention metric. Pre-training is employed to generate pseudo-labels for the target data, and Matching Filtering is applied to select high-quality labeled training pairs, enabling unsupervised training of the target domain. The model is evaluated using a comprehensive test set consisting of 34 subjects, 2 scenes, and 6 different outfits. The experimental results demonstrate the superior performance of DCS-Gait compared to existing methods, highlighting its effectiveness in cross-state and cross-scene gait identification tasks.

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